Unit - 1 {Thermodynamics}

Therme (Heat) Dynamics (Power)

Power = $\frac{W}{T}$

- Developed by Lord Kelvin
- At absolute temperature, thermal equillibrium occurs. Hence, K is the unit used.
- Thermodynamics is of 2 types :-

1. Statistical :- KTG (microscopic)
2. Classical :- Equillibrium (macroscopic)

- System ⇒ which is under investigation
- Surrounding ⇒ which is left except system.
- Boundary ⇒ can be real and imaginary.

→ Properties of system

- characteristics which are given some numerical values at a given time are called property of system
- They are of 2 types

1. Extensive properties. {depends on size} ⇒ m, V, K, electric charge
2. Intensive properties. {size independent} ⇒ P, T, ~~K~~, μ, density. ~~Density~~

Ratio of $\frac{\text{extensive}}{\text{extensive}}$ = Intensive.

State of thermodynamical system :-

⇒ Condition of the system as characterised by the system is called the state of the system. In other words, each unique condition of the system is called the state of system.

Equation of state of a thermodynamical system

⇒ $dU = Tds - PdV$ → Gibbs free energy eqⁿ.

Differentiate this w.r.t volume keeping T as const.

$$\frac{dU}{dV} = T\left(\frac{ds}{dV}\right) - P \rightarrow (2)$$

Accⁿ to Maxwell's eqⁿ :-

$$\left(\frac{ds}{dV}\right)_T = \left(\frac{dP}{dT}\right)_V \rightarrow (3)$$

$$\left(\frac{dU}{dV}\right)_T = T\left(\frac{dP}{dT}\right) - P \rightarrow (4)$$

1^{st} equation of state of a system → (4)

for ideal gaseous state

$PV = nRT$

for $n = 1$

$$P = \frac{RT}{V} \rightarrow (5)$$

differentiating (5) w.r.t temp with volume as constt.

$$\left(\frac{dP}{dT}\right)_V = \frac{R}{V} \rightarrow (6)$$

using (6) in (4)

$$\left(\frac{dU}{dV}\right)_T = T\left(\frac{R}{V}\right) - P$$

$$\left(\frac{dU}{dV}\right)_T = P - P$$

$$\left(\frac{dU}{dV}\right)_T = 0$$

Thermodynamical process

© A process is said to be occured if the system changes its state of eq^n from initial to some other state to eq^n or in such changes, system transformers its energy at eq^n from one state to another state in eq^n, then the process is occured.

Types of processes :-

in closed & open system

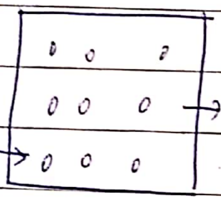
- ⇒ Isothermal → same temp
- ⇒ isobaric → iso stands for same.
- ⇒ isochoric
- ⇒ adiabatic → in isolated system
- ⇒ Reversible → in step by step
- ⇒ Irreversible → uncontrolled

5 eg of zeroth law :- thermometer

if u place a glass with hot H_2O & a glass with cold H_2O then they will attain eq^n with the temp of the room.

⊙ Concept of heat, internal energy and work

When heat given



→ state will change

changes observation by given heat

heat → PE, KE, work

KE → increases (change in internal energy)

$$W = F \times disp$$
$$= F \cancel{\times} A\, dl$$
$$\boxed{W = F \Delta V}$$

First law states that the total heat energy given is converted into the change in internal energy and the work done by the gas.

$dQ = dU + dW$

$\boxed{dQ = dU + P\Delta V}$

Applications of first law of thermodynamics.

① for cyclic process

$dQ = \cancel{dU}^{0} + \cancel{dU}^{0}\ dW$

$\boxed{dQ = dW}$

② for isothermal process

temp = const

$dU = 0$

$\boxed{dQ = dW}$

③ Adiabatic process

$\boxed{\Delta Q = 0}$

$\boxed{-dU = dW}$

④ Isochoric process

$\Delta V = 0$

$\boxed{dQ = dU}$

Work done in isothermal expansion

$\Rightarrow \quad dW = P\Delta V \quad \rightarrow (1)$

Integrating this

$$\int dW = \int_{V_1}^{V_2} P\,dV$$

$$W = P\int_{V_1}^{V_2} dv \quad \rightarrow (2)$$

As $PV = nRT$ {Real gas eqn}

$$\boxed{P = \frac{RT}{V}}$$

$$W = \int_{V_1}^{V_2} \frac{RT}{V}\,dV$$

$$\boxed{W = RT \log_e\left(\frac{V_2}{V_1}\right)}$$

$$\boxed{W = 2.303\ RT \log_{10} \frac{V_2}{V_1}}$$

$$\frac{V^{-\gamma+1}}{-\gamma+1}$$

Work done in adiabatic process $\Rightarrow dW = PdV \rightarrow ①$

$$W = \int_{V_1}^{V_2} PdV \rightarrow ②$$

As for adiabatic

$$PV^{\gamma} = K \quad , \quad \gamma = \frac{C_p}{C_v}$$

$$P = \frac{K}{V^{\gamma}} \rightarrow ③$$

$$W = \cancel{\int_{V_1}^{V_2} RT} \int_{V_1}^{V_2} \frac{K}{V^{\gamma}} dV$$

$$W = K \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2}$$

$$W = K \left[\frac{V_2^{-\gamma+1} - V_1^{-\gamma+1}}{-\gamma+1} \right]$$

Q. 5000 J of heat is added to 2 moles of an ideal monoatomic gas initially at temp of 500 K while gas performs 7500 J of work. What is the final temperature of gas.

$\cancel{Q = 5000}$ J $\quad n = 2$ moles

$\boxed{3/2 = \text{monoatomic gas}}$

$$dQ = \cancel{dW} \; dU + PdV$$

$$dQ = dU + \frac{nRT}{V} dV$$

SUNGRACE

$$5000 = dU + \frac{(2)(RT)}{V} dV = \boxed{dU = -2500}$$

CamScanner

Q a) Compute the internal energy change and temp change for 2 processes involving 1 mole of an ideal monoatomic gas with 1500 J heat is added to the gas and the gas does no work.

b) 1500 J of work is done on the gas and gas does no work and no heat is added or taken from the gas.

Calculate temperature.

Q. A 5 LBM system was taken from 50 °F to 150 °F. How much energy in the form of heat was added to the system to produce this temp increase. ($C_p = \frac{1.6 \times btu}{lbm - f}$)

Second law of thermodynamics

→ Natural law , $Q = u + w$

→ law of nature

EEE → Energy, Equillibrium, Entropy

↓

randomness of particle

Variations of entropy

solid (s) < liquid (l) < gas (g)

$\Delta S > 0$ (spontaneous process and it is irreversible)

$\Delta S = 0$ (reversible)

$\Delta S < 0$ (not possible)

→ Definitions

① Clausius statement :- The flow of heat from hot body to cold body is unadded, whereas the flow of heat from cold body to hot body is not possible without of help of external work.

② Kelvin plank statement ⇒ Its impossible to construct machine which convert total internal heat supply into useful work.
⇒

Second law in terms of entropy :- the entropy of the universe as an isolated system always ↑ everytime.

$$ds = \frac{dQ}{T} \quad \text{or} \quad S_2 - S_1 = \int_1^2 \frac{dQ}{T}$$

Entropy :- Measurement of randomness or disorder of a system is called entropy. It is ~~developed~~ denoted by S and mathematically defined as the change in entropy of a isolated system in reversible process.

$$\Delta S = + \frac{q_{rev}}{T}$$

Factors affecting entropy

① Physical state

$$S_s < S_L < S_g$$

② Temperature

(2) Temp

$T\uparrow$, $\Delta S \propto T$, $\Delta S \uparrow$, $KE\uparrow$, randomness $\uparrow$

(3) Pressure

$P\uparrow$, $\Delta S\downarrow$

$\Delta S \propto \frac{1}{P}$

(4) Mixing

$S + S = \Delta S > 0$

$S + L = \Delta S > 0$

$S + G = \Delta S > 0$

Properties of entropy

- Randomness or disorder
- Extensive (depends on size)
- Dynamic
- State function
- Unit = J/K or cal/K

Change in entropy for reversible process.

$dU = q + W$

$dU = q - W \rightarrow (1)$

$nC_v\,dt = q - W \rightarrow (2)$

$$q_{rev} = nC_v \Delta t + P_{ext} \Delta V \rightarrow (3)$$

$$q = nC_v \Delta T + P_{in} \Delta V \rightarrow (4)$$

$$\Delta S T = \frac{nC_v \Delta T + P_{in} \Delta V}{T}$$

$$\Delta S = \frac{nC_v \Delta T}{T} + \frac{P_{in} \Delta V}{T}$$

As per ideal gas equation

$$PV = nRT$$

$$P_{in} V = nRT$$

$$\frac{P_{in}}{T} = \frac{nR}{V} \rightarrow (6)$$

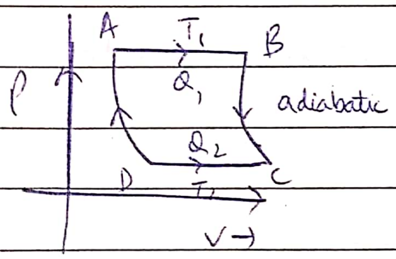
using (7) in (6)

$$\Delta S = \frac{nC_v \Delta T}{T} + \frac{nR \Delta V}{V}$$

by integrating

$$\Delta S_{\gamma} = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

graphically



for AB

$$\Delta S_1 = \frac{Q_1}{T_1}$$

for BC

$$\Delta S_2 = 0, \quad Q = \text{Const}$$

for CD

$\Delta S_3 = \frac{Q_2}{T_2}$

for DA

$\Delta S = 0$

by adding ① + ②

$$\Delta S_1 - \Delta S_2 = \frac{Q_1}{T_1} - \frac{Q_2}{T_2}$$

$$ds = 0$$

$$ds = ds_1 + ds_2 = \frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

for irreversible process

$$\eta_{ir} = 1 - \frac{Q_2}{Q_1}$$

$$\eta_r > \eta_{ir}$$

$$\eta_r = 1 - \frac{T_2}{T_1}$$

$$\Delta S_{ir} = \frac{Q_2}{Q_1} - \frac{T_2}{T_1} > 0$$

$$\left\{ \frac{Q_2}{T_2} > \frac{Q_1}{T_1} \right.$$

$$\Delta S_{ir} > 0$$

Thermodynamics

↓ Microscopic → ↓ statistical thermodynamics

↓ Macroscopic → ↓ Continuum approach → ↓ which explains the homogeneous nature of matter having less gap b/w the particles.

Continuum approach ⇒ According to continuum model, all the materials particles are exist or matter has the form of continuous homogeneous substance where there are no spaces or air gaps b/w the particles of the matter.

Eg. Water in bucket, air in container.

This model explains the stability of the matter as the size of the material particles increases, the probability of holes b/w the material particles will be low which tends to the matter into the stable form and the process is known as continuum approach.

Limitation of continuum process ⇒ The averaging in this model is ~~the~~ contradictory.

~~Eg.~~

Prove the relation :- $\oint P \cdot dV = \oint T \cdot ds$

Ans Using 1^{st} law and 2^{nd} law of thermodynamics.

$dQ = dU + dW$

$dQ = dU + P.dV \rightarrow (1)$

$ds = \frac{dQ}{T}$

$dQ = T ds \rightarrow (2)$

equating (1) = (2)

$T.ds = P.dV + \overset{0 \text{ (for cyclic process)}}{\cancel{dU}}$

Q. Compute the change in entropy when 1 g of ice at 0°C is converted into water at the same temp given that L.H.F is 80 cal/g.

$ds = \frac{dQ}{T}$

$ds = \cancel{80/1} \; \frac{dQ}{T}$

$\cancel{dQ = dU + dW}$

$\cancel{dQ = 336 \; \frac{80 \times 4.2}{10} + 1}$

$\cancel{\boxed{dQ = 336 + 1 = 337 \text{ Joules}}}$

$\cancel{ds = \frac{337}{273} = 1.23}$

$dQ = 80 \text{ cal/g}$

$\boxed{ds = \frac{80}{273}}$

Q. Calculate the heat required to convert 1 g of H_2O into steam under STP, if the entropy of H_2O at the B.P is 0.31 & for 100% steam at the steam temp is 1.749?

$PV = RT$

$dQ = T \cdot ds$

$Q = T(S_2 - S_1)$

$= 373(1.74 - 0.31)$

$Q = 373(1.43) = 533.39$ ~~kcal~~ cal/g

Q. 1 g molecules of perfect gas expands isothermally to 4 times its initial volume. Calculate change in entropy in terms of gas constant R.

$dQ = \int_{V_1}^{V_2} dW$ { isothermal }

$dQ = \int_{V}^{4V} P \cdot dV$

$dQ = \int_{V}^{4V} \frac{RT}{V} dV$

$\int dQ = RT \int_{V}^{4V} [\log(V)\, dV]$

$ds = \frac{164.34}{273}$

$Q = RT[\log(4V) - \log V]$

$Q = RT\left[\log_e\left(\frac{4\cancel{V}}{\cancel{V}}\right)\right]$

$\Rightarrow RT(\log_e 4)$

$\Rightarrow$ ~~$R(273)(0.6020)$~~

$\boxed{(RT)(\log_e 4)}$

~~$\boxed{dQ \Rightarrow 164.34R}$~~

$\log_e 4 = 2.303 \times \log 4$

$2.303 \times 2 \times \log 2 = \boxed{1.38}$

Q. for silver, the molar specific heat at const. P (C_P) in the range 50 - 100 K is given by $C_p = 0.076\,T - 0.00026\,T^2 - 0.15$ cal/K

where T is the Kelvin temp. If 2 mole of silver is heated from 50 - 100 K then calculate

① heat required
② change in entropy.

$$dQ = n C_p\, dt$$

$$\int dQ = 2\int_{50}^{100} C_p \cdot dt$$